

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 10 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. What is the mathematical formula used to determine the total number of rounds in a Single Knock-out tournament if the total number of participating teams (N) is exactly a power of 2 (e.g., 8, 16, 32)? (A) $N - 1$ (B) $\frac{N(N-1)}{2}$ (C) n (where $N = 2^n$) (D) $N + 1$ Unit 1: Management of Sporting Events | [Practice] | Tag: Very High**

Q2. Which essential micro-nutrient is a critical structural component of Vitamin B12, assisting in nerve function and blood formation? (A) Cobalt (Cobalamin) (B) Iodine (C) Sodium (D) Phosphorus Unit 5: Sports & Nutrition | [Practice] | Tag: Medium**

Q3. Which of the following yoga asanas is famously translated as the "Eagle Pose"? (A) Garudasana (Garun asana) (B) Dhanurasana (C) Tadasana (D) Makarasana Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: High**

Q4. The abbreviation 'CWSN' frequently used in physical education and inclusive schooling policies stands for: (A) Children With Special Needs (B) Children With Sensory Neuropathy (C) Care Without Special Needs (D) Children With Speech Needs Unit 4: Physical Education & Sports for CWSN | [Practice] | Tag: Very High**

Q5. Proteins are complex organic compounds crucial for muscle repair. They are formed by the combination of long chains of: (A) Fatty acids (B) Amino acids (C) Simple sugars (D) Glycerol Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High**

Q6. In the Rikli & Jones Senior Citizen Fitness Test battery, what is the standardized weight of the dumbbell used for MEN in the 'Arm Curl Test'? (A) 4 lbs (B) 5 lbs (C) 8 lbs (D) 10 lbs Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High**

Q7. A sharp, clean cut or wound through the skin and soft tissues caused by a sharp-edged object (like a broken glass piece or metal spike on the field) is medically termed as a/an: (A) Contusion (B) Laceration (C) Incision (D) Abrasion Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High**

Q8. In biomechanics, the SI unit used to measure 'Force' is the: (A) Joule (B) Watt (C) Newton (D) Kilogram Unit 8: Biomechanics and Sports | [Practice] | Tag: High**

Q9. The committee responsible for verifying the eligibility of teams, collecting entry forms, and drafting the initial tournament brackets is the: (A) Committee on Entries and Programmes (B) Announcement Committee (C) Logistics Committee (D) Decoration Committee Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High**

Q10. A person with a calculated Body Mass Index (BMI) of **22.5** falls strictly into which World Health Organization (WHO) category? (A) Underweight (B) Normal (Healthy) Weight (C)

Overweight (D) Obese Class I *Unit 5: Sports & Nutrition* | [PYQ] | Tag: Very High**

Q11. Which of the following is a rapid, forceful breathing technique (Pranayama) famously known as the "Bellows Breath" or "Skull Shining Breath", primarily used to cleanse the respiratory system? (A) Nadi Shodhana (B) Kapalbhathi (C) Sitli (D) Suryabhedan *Unit 3: Yoga as Preventive Measure for Lifestyle Disease* | [PYQ] | Tag: High**

Q12. The official logo of the Paralympic Games consists of three asymmetrical crescent shapes known as 'Agitos'. What are the three official colors of these Agitos? (A) Red, Blue, and Green (B) Red, Yellow, and Black (C) Blue, Black, and Green (D) Yellow, Green, and Red *Unit 4: Physical Education & Sports for CWSN* | [Practice] | Tag: Medium**

Q13. Which of the following non-nutritive components are chemical compounds produced by plants (e.g., flavonoids and anthocyanins) that help reduce the risk of cancer and cardiovascular diseases? (A) Artificial Sweeteners (B) Preservatives (C) Phytochemicals (D) Spices *Unit 5: Sports & Nutrition* | [PYQ] | Tag: Very High**

Q14. The Harvard Step Test uses a wooden bench of a specific height. What is the standard prescribed height of this stepping bench for MEN? (A) 16 inches (40 cm) (B) 18 inches (45 cm) (C) 20 inches (50 cm) (D) 24 inches (60 cm) *Unit 6: Test and Measurement in Sports* | [PYQ] | Tag: Very High**

Q15. In the immediate first-aid management acronym **PRICE** for soft tissue injuries, the letter 'C' specifically stands for: (A) Care (B) Cooling (C) Caution (D) Compression *Unit 7: Physiology & Injuries in Sport* | [PYQ] | Tag: Very High**

Q16. Assertion (A): Fluid friction (Air Resistance) significantly affects the flight trajectory of a projectile, such as a thrown javelin or discus. **Reason (R):** Air resistance is an opposing aerodynamic force that constantly slows down the horizontal velocity of the projectile during its flight. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports* | [Practice] | Tag: Very High**

Q17. Assertion (A): Knock-out tournaments are highly economical and save organizers a massive amount of time and resources. **Reason (R):** In a knock-out tournament, a team is permanently eliminated from the competition immediately after losing their very first match, leading to fewer total matches. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events* | [PYQ] | Tag: Very High**

Q18. Assertion (A): Artificial sweeteners are highly recommended for endurance athletes right before a competition to spike their energy levels. **Reason (R):** Artificial sweeteners perfectly duplicate the taste of sugar but actually contain zero or very low caloric energy. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition* | [PYQ] | Tag: High**

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Write down the exact mathematical formula used to calculate the total number of matches in a **Double League Tournament** and define what ' N ' represents in the formula.

*Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q20. 'Anorexia Nervosa' is a serious eating disorder frequently seen in aesthetic sports. Mention any two distinct symptoms or behaviors associated with it. *Unit 2: Children & Women in Sports | [Practice] | Tag: High***

Q21. Enlist any two medical contraindications (situations where it should be strictly avoided) for practicing *Vajrasana*. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q22. State any two physiological or psychological benefits that Children With Special Needs (CWSN) gain through active participation in adapted sports. *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: High***

Q23. Briefly describe the purpose and the specific scoring method of the 'Back Scratch Test' in the Senior Citizen Fitness Test battery. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q24. Differentiate clearly between an 'Incision' and a 'Laceration' as soft tissue sports injuries. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Medium***

SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

Q25. What do you mean by a 'Bye' in tournament fixtures? If 21 teams are participating in a single knock-out tournament, calculate the exact number of byes that must be allocated. Show your calculation. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q26. Elaborate on the physical exercise and screen-time guidelines recommended by the World Health Organization (WHO) specifically for adults (Age group 18-64 years). *Unit 2: Children & Women in Sports | [Practice] | Tag: High***

Q27. Explain the role of 'Phytochemicals' and 'Preservatives' as major non-nutritive components in modern dietary habits. *Unit 5: Sports & Nutrition | [PYQ] | Tag: High***

Q28. Detail the purpose, equipment required, and the step-by-step administration procedure of the '50m Dash (Standing Start)' test. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q29. What are the primary intrinsic and extrinsic causes of sports injuries? State any three general preventive measures athletes should take to avoid injuries on the field. *Unit 7: Physiology & Injuries in Sport | [Practice] | Tag: High***

Q30. Define a 'Projectile' in sports biomechanics. Discuss how the factors of 'Gravity' and 'Air Resistance' alter the parabolic flight path of a projectile. *Unit 8: Biomechanics and Sports* | [PYQ] | Tag: Very High**

SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

Q31. XYZ School is hosting an Inter-State Athletic Meet. The Principal has formed an Executive Committee which in turn created various sub-committees. During the actual sports event, a sprinter sprains his ankle on the track. Simultaneously, the chief guest arrives at the stadium gates, and the final results of the 100m sprint need to be announced over the loudspeakers. (i) Which specific committee is responsible for providing immediate medical treatment to the sprinter who sprained his ankle? (ii) The formal reception and welcoming of the chief guest at the stadium gates is the direct responsibility of the __ committee. (iii) Which committee is strictly responsible for using the public address system to inform the spectators and athletes about the 100m sprint results? (iv) Name the committee responsible for the registration of teams and preparing the fixtures before the tournament begins. **OR** Which committee handles media coverage, press releases, and sponsorships? *Unit 1: Management of Sporting Events* | [PYQ] | Tag: Very High**

Q32. The PE Teacher of a school organized a BMI (Body Mass Index) screening camp for Class 12 students. After analyzing the collected data, she categorized the students based on the standard WHO BMI norms. She noticed that students falling in the '> 30' category had a higher risk of lifestyle diseases, whereas students in the '< 18.5' category were physically weak and lethargic. (i) What is the mathematical formula used by the PE teacher to calculate the BMI of the students? (ii) The students falling in the '> 30' BMI category are diagnosed with which specific condition? (iii) What is the exact BMI numerical range that represents a 'Normal / Healthy Weight'? (iv) If a student is heavily obese, which macronutrient (that yields 9 kcal/g) should they significantly restrict in their diet? **OR** The students falling in the '< 18.5' category are classified as __. *Unit 5: Sports & Nutrition* | [PYQ] | Tag: Very High**

Q33. During a physics and physical education integrated seminar, a coach explains the mechanics of sports using video clips. He shows a clip of a sprinter aggressively driving his feet backward against the starting blocks to explode forward. He then shows a clip of a soccer ball rolling smoothly on a grass pitch until it eventually stops completely without hitting any obstacle. (i) The sprinter driving his feet backward into the blocks to propel his body forward is a perfect demonstration of Newton's _ Law of Motion. (ii) The rolling soccer ball eventually coming to a halt on the grass pitch is due to an invisible opposing force known as _. (iii) If the soccer ball was completely at rest on the penalty spot, it would not move until a player kicked it. This specific resistance to change in motion is termed as __. (iv) Define Newton's Second Law of Motion (Law of Acceleration). **OR** Differentiate briefly between sliding friction and rolling friction. *Unit 8: Biomechanics and Sports* | [PYQ] | Tag: Very High**

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. What is a 'League Tournament'? Thoroughly explain the 'Cyclic Method' of drawing league fixtures. Using this method, draw a neat fixture for 6 teams. Clearly display the number of matches and total rounds. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q35. Define 'Obesity' and state its major health hazards. Enlist any four Yoga Asanas prescribed in the CBSE syllabus for curing obesity. Thoroughly explain the step-by-step procedure, two physiological benefits, and two contraindications of *Dhanurasana* (Bow Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q36. The SAI Khelo India Fitness Test provides a comprehensive physical profile of students in schools. Elaborate in detail on the specific test items prescribed for the older age group (9 to 18 years / Classes 4 to 12) regarding the measurement of **Body Composition**, **Abdominal Core Strength**, and **Flexibility**. Give the procedure for each. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: High***

Q37. Biological limits dictate the maximum potential of an athlete. Discuss comprehensively the key physiological factors that determine an athlete's **Strength** and **Endurance** capacities (e.g., muscle fiber type, cardiac output, $\dot{V}O_2$ max, muscle cross-sectional area). *Unit 7: Physiology & Injuries in Sport | [PYQ 2024] | Tag: Very High***

Answer Key

Q1. (C) n (where $N = 2^n$) **Q2.** (A) Cobalt (Cobalamin) **Q3.** (A) Garudasana (Garun asana) **Q4.** (A) Children With Special Needs **Q5.** (B) Amino acids **Q6.** (C) 8 lbs **Q7.** (C) Incision **Q8.** (C) Newton **Q9.** (A) Committee on Entries and Programmes **Q10.** (B) Normal (Healthy) Weight **Q11.** (B) Kapalbhati **Q12.** (A) Red, Blue, and Green **Q13.** (C) Phytochemicals **Q14.** (C) 20 inches (50 cm) **Q15.** (D) Compression **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (D) (A) is false, but (R) is true. (Reason: Artificial sweeteners contain almost zero calories, so they *cannot* spike energy levels; they are non-nutritive).

Q19. Formula: Total Matches = $N(N - 1)$. **Definition:** ' N ' simply represents the total number of teams officially participating in the tournament. **Q20.** 1. Intense and irrational fear of gaining weight, leading to extreme starvation or skipping meals. 2. Distorted body image (believing they are overweight even when severely underweight or malnourished). **Q21.** 1. Individuals suffering from chronic knee pain, acute joint injuries, or severe arthritis should avoid it. 2. People with hernia or severe intestinal ulcers must strictly not perform this asana. **Q22.** 1. **Physical:** Enhances gross motor skills, cardiovascular endurance, and coordination. 2. **Psychological:** Massively boosts self-esteem, reduces depression, and develops a sense of independence. **Q23. Purpose:** To assess the flexibility of the upper body (specifically the

shoulders), which is essential for reaching tasks like combing hair. **Scoring:** The overlap (or gap) between the middle fingers behind the back is measured in inches or centimeters. Zero is touching, positive is overlapping, negative is a gap. **Q24.** An **Incision** is a sharp, clean-cut wound caused by a sharp-edged object (like broken glass), resulting in straight, smooth wound edges. A **Laceration** is an irregular, jagged tear in the skin caused by a blunt force trauma or a heavy, rough impact.

Q25. A **Bye** is a privilege given to a team (usually drawn by lots) in a knock-out fixture, which directly exempts them from playing in the first round and advances them to the second round. **Calculation for 21 teams:** * Total Teams (N) = 21. * Next highest power of 2 = 32 (2^5). * Number of Byes = Next power of 2 minus N * Byes = $32 - 21 = 11$ Byes. **Q26. WHO**

Guidelines for Adults (18-64 yrs): 1. Should engage in at least 150-300 minutes of moderate-intensity aerobic physical activity, OR 75-150 minutes of vigorous-intensity activity per week.

2. Muscle-strengthening activities involving major muscle groups should be done on 2 or more days a week. 3. Screen-time and prolonged sedentary behavior must be actively replaced with physical movement of any intensity. **Q27. Phytochemicals:** Plant-based chemical compounds (like anthocyanins in berries or flavonoids in beans) that are non-nutritive but act as powerful antioxidants, protecting the body against cardiovascular diseases and thwarting cancer cells. **Preservatives:** Synthetic or natural chemical additives mixed into foods and beverages to inhibit microbial/bacterial growth, preventing spoilage and greatly extending the food's shelf life. **Q28. Purpose:** To evaluate maximum acceleration and explosive speed. **Equipment:** Measuring tape, stopwatch, and cones. **Procedure:** A 50-meter straight track is marked. The athlete stands behind the starting line. On the command "Ready... Go!", the athlete sprints as fast as possible across the finish line. The time taken to cover the exact 50m distance is recorded in seconds. **Q29. Causes:** Poor physical conditioning, lack of warm-up, faulty equipment, reckless aggression, or overtraining.

Preventive Measures: 1. *Proper Warm-up:* Must engage in general and specific warm-ups to increase muscle temperature before play. 2. *Use of Protective Gear:* Always wear mandatory safety equipment like shin guards, helmets, or gum guards. 3. *Fair Play:* Strictly abide by the rules and avoid malicious fouls or reckless tackles. **Q30.** A **Projectile** is any object (or athlete's body) launched into the air that moves under the sole influence of gravity and air resistance (e.g., a thrown shot put). **Gravity:** A constant downward force pulling the projectile back to the earth, creating the downward curve of the parabola. **Air Resistance (Fluid Friction):** An opposing aerodynamic force that constantly pushes against the projectile's motion, decreasing its horizontal velocity and shortening its flight distance.

Q31. (i) First Aid Committee (or Medical Committee). (ii) Reception Committee. (iii) Announcement Committee. (iv) Committee on Entries and Programmes (or Registration/Technical Committee). **OR** Marketing (or Publicity/Sponsorship) Committee. **Q32.** (i) $BMI = \frac{\text{Weight (in kg)}}{\text{Height (in meters)}^2}$. (ii) Obesity (Class I, II, or III depending on exact range). (iii) 18.5 to 24.9. (iv) Fats (Lipids). **OR** Underweight. **Q33.** (i) Third Law (Action and Reaction). (ii) Rolling Friction. (iii) Inertia. (iv) Newton's Second Law states that the rate of change of momentum (acceleration) of an object is directly proportional to the applied force and occurs in the

direction of the force ($F = ma$). **OR** Sliding friction acts when two surfaces slide over one another (ice skating), whereas rolling friction acts when an object rolls over a surface (a rolling football). Rolling friction is much weaker than sliding friction.

Q34. League Tournament: A format (also called Round Robin) where every participating team plays against every other team at least once, regardless of winning or losing previous matches. It decides the true winner based on points accumulated. **Calculations for 6 Teams**

($N = 6$, Even Number): * Total Matches = $\frac{N(N-1)}{2} = \frac{6 \times 5}{2} = 15$ matches. * Total Rounds = $N - 1 = 6 - 1 = 5$ rounds. **Cyclic Fixture Draw:** (In an even number of teams, Team '1' is fixed at the top right. Other teams rotate clockwise). * Round 1: 1-6, 2-5, 3-4 * Round 2: 1-5, 6-4, 2-3 * Round 3: 1-4, 5-3, 6-2 * Round 4: 1-3, 4-2, 5-6 * Round 5: 1-2, 3-6, 4-5 (*Evaluator note: Full marks for showing Team 1 static and correctly rotating others clockwise across 5 rounds*). **Q35.**

Obesity is an abnormal or excessive accumulation of body fat ($BMI \geq 30$) that severely impairs health, leading to diabetes, hypertension, and heart attacks. **Asanas:** Vajrasana, Hastotanasana, Trikonasana, Ardha Matsyendrasana, Dhanurasana. **Dhanurasana (Bow Pose) Procedure:** Lie prone (on the stomach). Bend both knees bringing the heels close to the buttocks. Reach back with both hands and firmly grasp the ankles. Inhale deeply, pull the legs upward and backward, simultaneously lifting the chest and head off the floor, balancing the entire body weight on the abdomen. Hold, exhale, and release. **Benefits:** 1. Massively stretches abdominal muscles, melting away belly fat. 2. Cures postural deformities of the spine like kyphosis by expanding the chest. **Contraindications:** 1. Avoid if suffering from hernia or severe lower back pain/slipped disc. 2. Pregnant women or individuals with recent abdominal surgeries must not perform it. **Q36.** The SAI Khelo India battery measures overall physical fitness using specific standardized protocols for the 9-18 years age group: 1. **Body Composition (BMI):** Evaluates if the child's weight is healthy. *Procedure:* The child's body weight is measured in kilograms using a digital scale. The height is measured in meters using a stadiometer. BMI is calculated as $Weight/Height^2$. 2. **Abdominal/Core Strength (Partial Curl-up):** *Procedure:* The student lies supine on a flat, cushioned surface with knees flexed at 90° and feet flat. Arms are straight at the sides, palms down. On command, the subject raises their trunk smoothly, sliding fingers forward exactly 6 inches along the floor, then returns until the shoulder blades touch the mat. *Scoring:* Maximum correct curl-ups performed in exactly 30 seconds. 3. **Flexibility (Sit and Reach Test):** *Procedure:* The student sits barefoot with legs fully extended, feet flat against a wooden testing box. They slowly reach forward with both hands sliding along the scale as far as possible and hold the maximum stretch for 2 seconds. *Scoring:* Maximum distance reached recorded in centimeters. **Q37. Physiological Factors Determining Strength:** 1. **Muscle Cross-Sectional Area:** The larger the physical thickness/girth of the muscle (hypertrophy), the more contractile protein it possesses, directly resulting in greater maximum force production. 2. **Muscle Fiber Composition:** Muscles contain genetically determined ratios of fast-twitch (white) and slow-twitch (red) fibers. A higher proportion of fast-twitch fibers allows the muscle to produce massive, explosive force rapidly, essential for weightlifters. 3. **Motor Unit Recruitment:** Strength relies on the central nervous system's ability to simultaneously recruit a larger number of motor units to contract the muscle efficiently. **Physiological Factors Determining Endurance:** 1. **Maximum Oxygen**

Uptake ($\dot{V}O_2$ Max): The highest amount of oxygen the body can absorb, transport, and utilize in one minute. A higher $\dot{V}O_2$ max means oxygen reaches working muscles faster, massively boosting aerobic endurance. 2. **Cardiac Output & Stroke Volume:** A highly trained heart pumps a massive volume of oxygen-rich blood per beat (stroke volume) and per minute (cardiac output), drastically delaying muscular fatigue. 3. **Lactic Acid Tolerance:** During intense exercise, lactic acid accumulates, causing burning and fatigue. Elite endurance athletes physiologically adapt to tolerate and clear higher levels of lactic acid, allowing them to sustain high-intensity work longer.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.